

Chapters 19 and 20: Basis, Coordinates, and Dimension Worksheet

1. Are each of the following statements true or false?

- (a) The space spanned by $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix} \right\}$ has dimension 2.
- (b) If U is a subspace of a vector space V with $\dim U = 1$, then $U = \{c\mathbf{x}\}$ for some nonzero vector $\mathbf{x}$, where c is in $\mathbb{R}$.
- (c) Every basis of $M_{2 \times 2}$ contains a noninvertible matrix.
- (d) A set of 4 polynomials in P_2 will be linearly dependent.
- (e) $M_{3 \times 4}$, $M_{4 \times 3}$, $M_{6 \times 2}$, and P_{11} all have the same dimension.
- (f) Suppose U is a vector space of dimension n , then any $n + 1$ vectors of U are linearly dependent.
- (g) $\mathbb{R}^2$ has infinitely many vectors, so it is infinite-dimensional.
- (h) Any four polynomials in P_3 will form a basis for P_3 .

Solution.

- (a) False, these vectors are linearly dependent, so they don't form a basis for their span.
- (b) True.
- (c) False. For example: $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right\}$.
- (d) True. P_2 has dimension 3, so if a set of 4 polynomials in P_2 were linearly independent, they would span a 4-dimensional subspace of a 3-dimensional vector space, which is impossible.
- (e) True. They're all dimension 12.
- (f) True.
- (g) False, the dimension of a vector space is the number of vectors in a BASIS for that vector space. The dimension of $\mathbb{R}^2$ is 2.
- (h) False. A basis for P_3 needs four polynomials, but the polynomials need to be linearly independent to be a basis.

2. Find a basis for the solution space of $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 & 2 \\ -1 & 0 & -9 & 9 & 4 \\ 1 & 3 & 0 & 5 & -2 \\ 2 & 4 & 6 & -3 & 2 \end{bmatrix}, \quad RREF(A) = \begin{bmatrix} 1 & 0 & 9 & 0 & -22 \\ 0 & 1 & -3 & 0 & 10 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

What's the dimension of this space?

Solution.

The solution space of $A\mathbf{x} = \mathbf{0}$ lives in $\mathbb{R}^5$: $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$, where x_3, x_5 are free variables. So I set

$x_3 = r, x_5 = t$. We then solve for the remaining variables: $x_4 = 2t, x_2 = 3r - 10t$, and

$$x_1 = -9r + 22t. \text{ So our vectors look like } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -9r + 22t \\ 3r - 10t \\ r \\ 2t \\ t \end{bmatrix} = r \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 22 \\ -10 \\ 0 \\ 2 \\ 1 \end{bmatrix}. \text{ So our}$$

spanning set is $\begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 22 \\ -10 \\ 0 \\ 2 \\ 1 \end{bmatrix}$. This set is also linearly independent (in the linear combination

above, both $r = 0$ and $t = 0$ to make the zero vector), so this is a basis of the solution space of $A\mathbf{x} = \mathbf{0}$. Since there are two vectors in a basis, the dimension of the solution space of $A\mathbf{x} = \mathbf{0}$ is 2.

3. Given a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$, where:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Find a basis for the subspace spanned by S . (Hint: You did a lot of the work for this problem on a previous worksheet).

Solution.

We showed that $\mathbf{v}_3$ lies in the span of $\mathbf{v}_1, \mathbf{v}_2$, so $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$. If that's unclear to you, note that every linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3$ can be replaced with a linear combination of $\mathbf{v}_1, \mathbf{v}_2$, $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3(2\mathbf{v}_2 - \mathbf{v}_1)$. To see that $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent, we just chop off the third column of our original matrix:

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

This has no free variables, so $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent (also, they're not scalar multiples

of each other. So our basis is: $\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ I know what you're thinking: Why couldn't we use $\mathbf{v}_3$

and one of the other vectors? Sure, that also works. HOWEVER, because of the order of the vectors in our matrix, it makes sense to use $\mathbf{v}_1, \mathbf{v}_2$, because we can clearly see that they're linearly independent (they both correspond to pivots). This will set up our algorithm for finding a basis from a spanning set.

4. Consider the set $S = \{x^2 - 3x + 1, 4x - 2x^2 + 3, 5 - 2x\}$

(a) Does $p(x) = x^2 - 3x + 2$ lie in span S ?

(b) Does S form a basis for P_2 ?

Solution.

(a) We're looking for a, b, c with

$a(x^2 - 3x + 1) + b(4x - 2x^2 + 3) + c(5 - 2x) = x^2 - 3x + 2$. Collecting like terms, this means $a - 2b = 1, -3a + 4b - 2c = -3, a + 3b + 5c = 2$. We put this into an augmented matrix and reduce:

$$\begin{bmatrix} 1 & -2 & 0 & 1 \\ -3 & 4 & -2 & -3 \\ 1 & 3 & 5 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The bottom row tells us this is an inconsistent system, so $p(x)$ does not lie in the span of S .

(b) S does not form a basis for P_2 . We can tell from part (1): Because $p(x)$ does not lie in the span of S , S does not span P_2 , so S is not a basis for P_2 .

Notice if we ignore the last column of our augmented matrix, we can also see that this set is not linearly independent (since the last row would be all zeros, so we'd have free variables in the homogeneous system), which is another reason it's not a basis for P_2 .

5. Consider the subset W of M_{22} where $W = \left\{ A \mid A \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} A \right\}$.

- (a) Show that W is a subspace of M_{22} .
- (b) Find a basis for W .
- (c) What is the dimension of W ?

Solution.

- (a) I'm going to put in a place-holder for $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = X$ (you'll see that the specific matrix X doesn't matter here). Let's show this is a subspace:

Non-empty: The zero matrix O is in W since $OX = O = XO$.

Closed under addition: Suppose A, B are in W . Then $AX = XA$ and $BX = XB$. Let's consider $(A + B)$:

$(A + B)X = AX + BX = XA + XB = X(A + B)$, so $(A + B)$ is in W .

Closed under scalar multiplication: Let A be in W , c in $\mathbb{R}$. Then $AX = XA$. Let's consider cA :

$(cA)X = cAX = cXA = X(cA)$ (remember that scalars can move around a matrix product), so cA is in W .

- (b) Now we need to use $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$. I'm going to write $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and perform the matrix multiplication:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} a - b & a \\ c - d & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a + c & b + d \\ -a & -b \end{bmatrix}$$

Equating the two, we have four equations:

$a - b = a + c, a = b + d, c - d = -a, -b = c$, which can be rearranged into four homogeneous equations: $b + c = 0, a - b - d = 0, a + c - d = 0, b + c = 0$. We can set up a matrix and reduce:

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & -1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Set $c = r, d = t$, the solve: $a = t - r, b = -r$, so our matrices have the form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} t - r & -r \\ r & t \end{bmatrix} = r \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} + t \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

It's easy to see that $\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ span this set, and are linearly independent (they're not scalar multiples of each other AND you can see that a linear combination of them = O forces $r = 0$ and $t = 0$), so this is a basis for W .

- (c) There are two vectors in the basis, so the dimension is 2.

6. If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis of V , which of the following are bases? Which of them can span V ?

- (a) $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$.
- (b) $\{2\mathbf{u} + \mathbf{v} + 3\mathbf{w}, 3\mathbf{u} + \mathbf{v} - \mathbf{w}, \mathbf{u} - 4\mathbf{w}\}$.
- (c) $\{\mathbf{u}, \mathbf{u} + \mathbf{v} + \mathbf{w}\}$.
- (d) $\{\mathbf{u}, \mathbf{u} + \mathbf{w}, \mathbf{u} - \mathbf{w}, \mathbf{v} + \mathbf{w}\}$.

Solution.

- (a) $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$ is also linearly independent. We can directly test $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$ for linear independence:
 $t_1(\mathbf{u} + \mathbf{v}) + t_2(\mathbf{v} + \mathbf{w}) + t_3(\mathbf{w} + \mathbf{u}) = \mathbf{0}$, so $(t_1 + t_3)\mathbf{u} + (t_1 + t_2)\mathbf{v} + (t_2 + t_3)\mathbf{w} = \mathbf{0}$.
Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, this means $t_1 + t_3 = 0, t_1 + t_2 = 0, t_2 + t_3 = 0$
and solving this we get that $t_1 = t_2 = t_3 = 0$. Thus, $\{\mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{w} + \mathbf{u}\}$ is also a basis for V .

- (b) $\{2\mathbf{u} + \mathbf{v} + 3\mathbf{w}, 3\mathbf{u} + \mathbf{v} - \mathbf{w}, \mathbf{u} - 4\mathbf{w}\}$ is not linearly independent.

Similar to previous solution: $t_1(2\mathbf{u} + \mathbf{v} + 3\mathbf{w}) + t_2(3\mathbf{u} + \mathbf{v} - \mathbf{w}) + t_3(\mathbf{u} - 4\mathbf{w}) = \mathbf{0}$, so
 $(2t_1 + 3t_2 + t_3)\mathbf{u} + (t_1 + t_2)\mathbf{v} + (3t_1 - t_2 - 4t_3)\mathbf{w} = \mathbf{0}$. Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent, this means $2t_1 + 3t_2 + t_3 = 0, t_1 + t_2 = 0, 3t_1 - t_2 - 4t_3 = 0$. However, this system does have a nontrivial solution (ex: $t_1 = 1, t_2 = -1, t_3 = 1$), so this set is NOT linearly independent, thus not a basis for V . Since V has dimension 3, three linearly dependent vectors cannot span V , so this set doesn't span V .

- (c) $\{\mathbf{u}, \mathbf{u} + \mathbf{v} + \mathbf{w}\}$ can't be a basis for V as V must have dimension 3. Since there are only two vectors, it cannot span V .
- (d) $\{\mathbf{u}, \mathbf{u} + \mathbf{w}, \mathbf{u} - \mathbf{w}, \mathbf{v} + \mathbf{w}\}$ can't be a basis for V as V must have dimension 3. Since there are 4 vectors, the set must be linearly dependent.

However, since

$\mathbf{u} = 1\mathbf{u}, \mathbf{w} = \frac{1}{2}(\mathbf{u} + \mathbf{w}) - \frac{1}{2}(\mathbf{u} - \mathbf{w}), \mathbf{v} = 1(\mathbf{v} + \mathbf{w}) - \frac{1}{2}(\mathbf{u} + \mathbf{w}) + \frac{1}{2}(\mathbf{u} - \mathbf{w})$, each of our basis vectors are in the span of this set, so it must also span V .

7. Consider $S = \{x, x + 1, x^2 + 2\}$ and $B = (x, x + 1, x^2 + 2)$.

(a) Show that S is a basis for P_2 .

(b) If the coordinates for q relative to the ordered basis B are $[q]_B = C_B(q) = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, find q .

(c) Using ordered basis B , find the coordinate vector of $p = 3x - 3x^2 - 8$ relative to B .

Solution.

(a) Since P_2 is 3-dimensional, and there are 3 polynomials in S , it's sufficient to check that S spans P_2 or S is linearly Independent. Let's check linearly independent: We form a linear combination of S and set it equal to 0: $a_1(x) + a_2(x + 1) + a_3(x^2 + 2) = 0$. Collecting like terms, we find that $a_1 + a_2 = 0$, $a_2 + 2a_3 = 0$, and $a_3 = 0$. $a_3 = 0$ means that $a_2 = -2a_3 = 0$, and this forces $a_1 = -a_2 = 0$. So this set is linearly independent, which makes it a basis for P_2 .

(b) $[q]_B = C_B(q) = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ gives us the coefficients for the linear combination using the

basis B : $q = 2(x) + 5(x + 1) - 1(x^2 + 2) = -x^2 + 7x + 3$.

(c) We now want to find a_1, a_2, a_3 such that

$$a_1(x) + a_2(x + 1) + a_3(x^2 + 2) = 3x - 3x^2 - 8 = p.$$

Collecting like terms, we find that $a_1 + a_2 = 3$, $a_2 + 2a_3 = -8$, and $a_3 = -3$. $a_3 = -3$ means that $a_2 = -8 - 2(-3) = -2$, and this implies $a_1 = 3 - (-2) = 5$. Our

coordinate vector is then $[p]_B = C_B(p) = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$ (note that the order follows the order

given in B).